



Sensors and Instrumentation

Smart Greenhouse Monitor and Controller

Academic year: 2025-2026

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Contents

1. Introduction	3
2. Problem Statement	4
3. System Architecture	4
4. Sensor Selection and Justification	8
5. Hardware Design and Schematics	9
a) Hardware design	9
b) Connection and Pinout Mapping	9
c) Opto-isolated Relay Interface and High-Current Control	10

d)	Functional Operation	11
e)	Load Connections (Normally Open Configuration)	11
f)	Schematic Representation.....	12
6.	Software Design and Firmware Architecture.....	12
a)	Programming Environment.....	12
b)	Modular Structure	13
c)	Sampling Rates.....	13
d)	Calibration Constants	13
e)	Serial Output	14
f)	Control Logic	14
g)	Flowchart	14
7.	Calibration Procedures and Results	14
i.	Soil Moisture sensor	14
a)	Raw Results.....	15
b)	Reference values	15
c)	Calibration Equation	15
ii.	Light Sensor (LDR):	18
a)	The code logic is	18
b)	Raw values	18
c)	Calibration Curve Equation	19
iii.	Temperature and Humidity Sensor (DHT11).....	21
iv.	Flow Sensor (YF-S201)	25
v.	Weight Sensor (HX711 + Load Cell).....	27
8.	Experimental Results and Data Analysis.....	30
i.	Light Sensor (LDR):.....	30
a)	Data Table	30
b)	Results.....	31
c)	Data Analysis.....	31
ii.	Temperature and Humidity Sensor (DHT11)	31
a)	Data Table	31
b)	Results.....	32
c)	Data Analysis.....	32
iii.	Flow sensor (YF-S201)	33

a)	Data Tables.....	33
b)	Results.....	34
c)	Data Analysis.....	35
iv.	Soil Moisture and Weight (HX711 + Load Cell) sensor.....	36
a)	Data Tables.....	36
b)	Results.....	36
c)	Data Analysis.....	37
9.	Measurement Uncertainty and Error Analysis	38
i.	Sensor Accuracy and Resolution.....	38
ii.	Repeatability	39
iii.	Possible Improvements to Measurement Quality	39
10.	Engineering Constraints Discussion	40
i.	Cost Considerations.....	40
ii.	Reliability and Robustness	40
iii.	Electrical Safety	40
iv.	Maintainability and Power Consumption	41
v.	Scalability and Future Improvements	41
11.	Conclusion and Future Work	41
12.	References	42

1. Introduction

Automation is increasingly integrated in modern agricultural systems in order to ensure optimal plant growth while resource waste is reduced. This can be achieved by the means of greenhouses that control the environment. However, the manual monitoring of light, temperature and humidity may result in inefficiencies like inconsistent irrigation, poor light management and resource waste.

This project is an Arduino-based Smart Greenhouse Monitor and Controller that measures five key environmental parameters:

1. Light intensity
2. Soil weight
3. Water flow
4. Temperature

5. Humidity

Consequently, it automatically actuates a grow lamp, pump and ventilation fan-based on the threshold logic. Although the sensors and microcontrollers in this system are low cost, but they can achieve the greenhouse control principles and support precision in agriculture.

2. Problem Statement

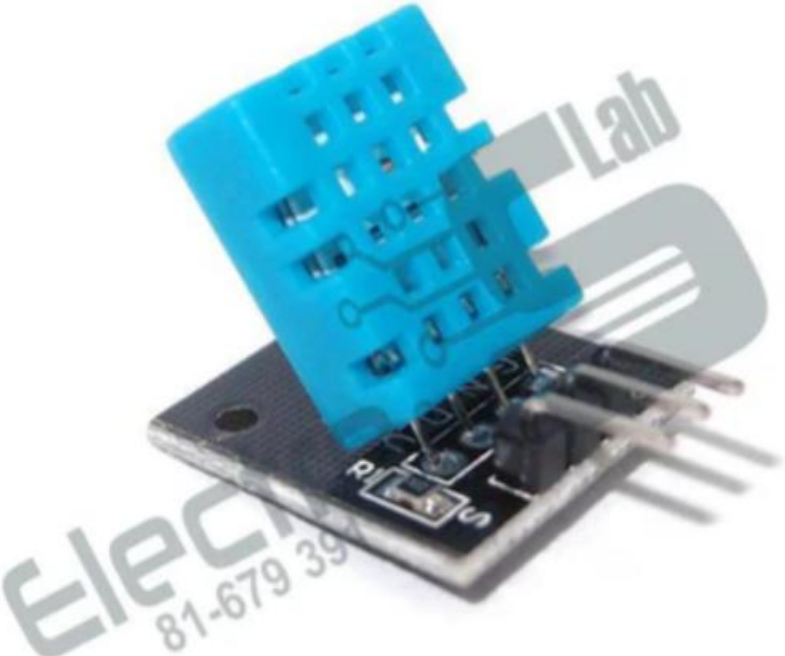
Human observation and manual control are the traditional management technique used in greenhouses and they can result in inconsistent watering, lightning, and ventilation.

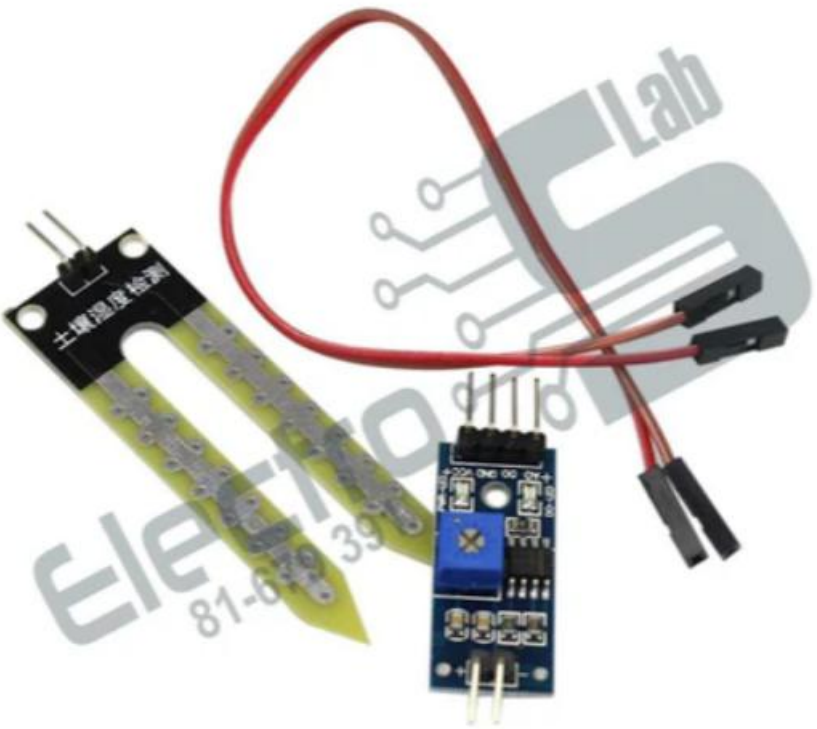
The problem to be addressed is how to maintain optimal and healthy growth conditions automatically using real time sensor feedback with a reliable and low-cost embedded system that monitors the variables and takes actions when thresholds are exceeded.

3. System Architecture

This system is built using 5 sensors in addition to other components.

Subsystem	Components	Function
Optical sensor	Light Sensor LDR 5mm 	Measures ambient light to control grow lamp
Water flow sensor	YF-S201 	Measures irrigation water flow rate
Temperature and Humidity sensor	DHT11	Measures air humidity

Subsystem	Components	Function
		for ventilation control
Electronic weighting sensor kit		Measures the weight of plant pots

Subsystem	Components	Function
Soil moisture sensor module		It detects the level of water in soil
Actuation	2-Channel Relay Module 	Drives led light, pump, and fan. It acts as an electrically isolated switch to safely allow the low voltage Arduino to control higher voltage and higher current components



In addition to these components, many other components are used such as:

- ❖ LCD Screen: to display the output of every sensor

- ❖ Arduino UNO: It's the central brain that's responsible for reading the sensor data and controlling all actuators. It's an easy-programmable circuit that offers sufficient pins for input/output port.



- ❖ Submersible pump: for water supply

- ❖ 1m pipe for small water pump



- ❖ Cooling fan: It regulates the greenhouse temperature and improves air circulation.



- ❖ Led Light
- ❖ 12V adapter for the pump
- ❖ 5V adapter for the fan
- ❖ Tank for water
- ❖ A plant

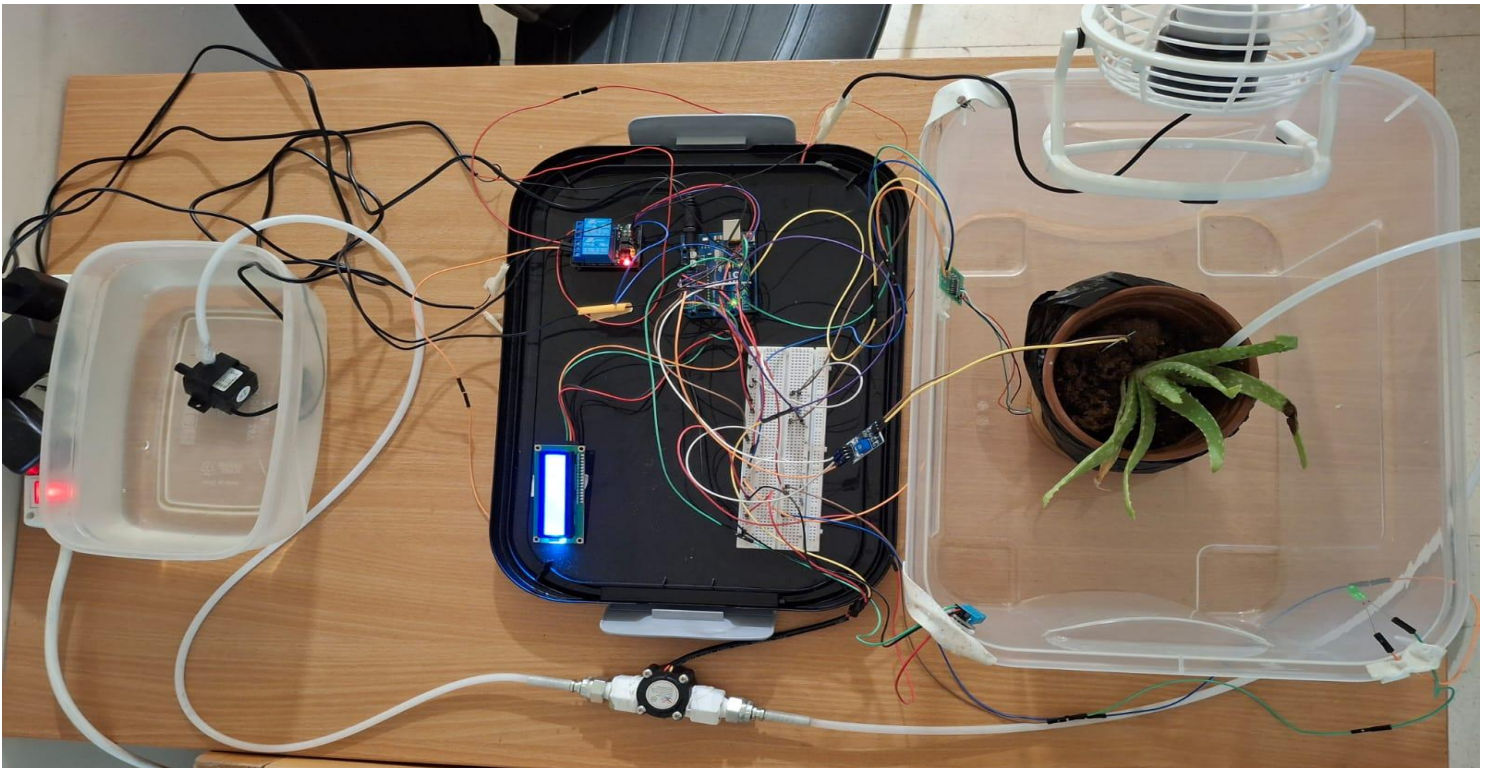
4. Sensor Selection and Justification

- As an **optical sensor**, we used the **LDR (Light dependent resistor)** to monitor ambient light intensity. It's low-cost and simple to use and it's more preferable for such application. Since plants require adequate light for photosynthesis, the sensor allows the system to detect low-light conditions and automatically switch on the LED grow light when natural sunlight is insufficient.
- For **temperature and humidity sensor**: The **DHT11** sensor has been selected. Temperature and relative humidity are two factors that affect the growth of the plant and measuring both of them via a single sensor can be achieved by the DHT11 sensor. Hence, we can reduce system complexity and cost.
- For **soil moisture sensor**, we selected a soil moisture sensor to continuously monitor the water content and soil because both overwatering and underwatering can damage plants. Integrating the use of this sensor in this project would lead to an automatic activation of the irrigation pump when necessary, and deactivation to improve water efficiency.
- **Electronic weighting sensor kit (Load Cell+HX711)**: is selected to monitor the weight of the plant pot that when decreases it directly indicates water loss due to evaporation. Using this sensor can be considered as an additional method for evaluating irrigation needs, increasing the reliability of the whole system.

- *YF-S201 water flow sensor* was selected to measure the amount of water delivered by the pump during irrigation. It can detect the malfunctioning in pumps and blocked pipes and ensures that the plant receives sufficient amount of water.

5. Hardware Design and Schematics

a) Hardware design



The physical implementation of the automated greenhouse system relies on an Arduino microcontroller acting as the central processing unit. The hardware design is structured to handle various sensor inputs (analog and digital) and low-voltage control signals, while safely isolating and switching the higher-current inductive loads (the DC water pump and the ventilation fan) using an opto-isolated relay module.

b) Connection and Pinout Mapping

To ensure reliable, non-conflicting communication between the microcontroller and the peripheral hardware, a strict pin allocation scheme was established. The table below details the complete wiring interconnectivity of the system:

Component	Component Pin	Arduino Pin	Signal Type	Description / Function
YF-S201 Flow Sensor	Out (Yellow)	Pin 2	Digital Input (Interrupt)	Measures water flow volume via pulse count
2-Ch Relay Module	IN2 (Pump Control)	Pin 4	Digital Output	Controls the top relay channel to switch the 12V water pump
2-Ch Relay Module	IN1 (Fan Control)	Pin 5	Digital Output	Controls the bottom relay channel to switch the ventilation fan
Soil Moisture Sensor	VCC	Pin 6	Digital Output	Switched VCC power line to prevent probe corrosion
DHT11 Sensor	Data	Pin 7	Digital Input	Reads ambient temperature and humidity data
HX711 Load Cell	DOUT	Pin 8	Digital Input	Serial data output line for weight measurement
HX711 Load Cell	PD_SCK	Pin 9	Digital Output	Power-down and serial clock input line
Grow Light LED	Anode (+)	Pin 10	Digital Output	Powers the supplemental LED grow light
Soil Moisture Sensor	Analog Out (A0)	Pin A0	Analog Input	Reads moisture level (mapped 0% to 100%)
LDR (Photoresistor)	Analog Out (A0)	Pin A3	Analog Input	Measures ambient light intensity
LCD 16x2 (I2C)	SDA / SCL	SDA / SCL (A4/A5)	Serial Data / Clock	Displays real-time telemetry

c) Opto-isolated Relay Interface and High-Current Control

A key engineering challenge in this project is safely controlling the 12V DC water pump and the ventilation fan using the Arduino's low-power 5V logic outputs. Direct connection is impossible, as inductive motor loads draw transient currents that would instantly destroy the microcontroller's digital pins.

To solve this, a Songle SRD-05VDC-SL-C 2-Channel Relay Module is integrated into the system:

d) Functional Operation

- **Opto-isolation (PC817 Optocouplers):** The module features onboard optocouplers (the small black 4-pin ICs behind the input header). When the Arduino pin drives a channel LOW, current flows through an internal LED on the relay board. This light triggers a phototransistor on the isolated side, completing the circuit to energize the electromagnetic relay coil. Because there is **no physical electrical connection** between the Arduino's logic circuit and the relay's switching circuit, the microcontroller is entirely protected from electromagnetic interference (EMI) and voltage spikes.
- **Relay Coil and Flyback Protection:** The SONGLE SRD-05VDC-SL-C relays require a 5V supply to energize their internal coils. Flyback diodes (D1 and D2) are placed in parallel with the coils to safely dissipate high-voltage inductive spikes generated when the magnetic field collapses upon turn-off.
- **Common Ground / JD-VCC Jumper:** The yellow jumper on the right side of the module (connecting VCC and JD-VCC) allows the relay coils to share the Arduino's 5V rail. For absolute physical isolation, this jumper can be removed, and an external auxiliary 5V supply can be connected directly to JD-VCC and GND.

e) Load Connections (Normally Open Configuration)

The high-power circuits (the 12V pump and fan) are wired to the blue screw terminal blocks on the left side of the relay module. Each relay channel acts as a single-pole double-throw (SPDT) switch containing three terminals: Common (COM), Normally Open (NO), and Normally Closed (NC).

To ensure safety, both the pump and the fan are wired in a Normally Open (NO) configuration:

1. The positive wire of the external 12V DC power supply connects to the COM terminal.
 2. The positive terminal of the load (pump or fan) connects to the NO terminal.
 3. The negative wire of the load connects directly back to the power supply's ground (forming a complete loop when closed).
- **Software Logic:** Because this relay module operates on active-low logic, writing a LOW signal from the Arduino (digital Write (pin, LOW)) energizes the coil, closing the contact between COM and NO, thereby turning the load ON. Conversely, writing a HIGH signal or system power loss keeps the relay de-energized, leaving the circuit Open and the loads safely OFF.

f) Schematic Representation

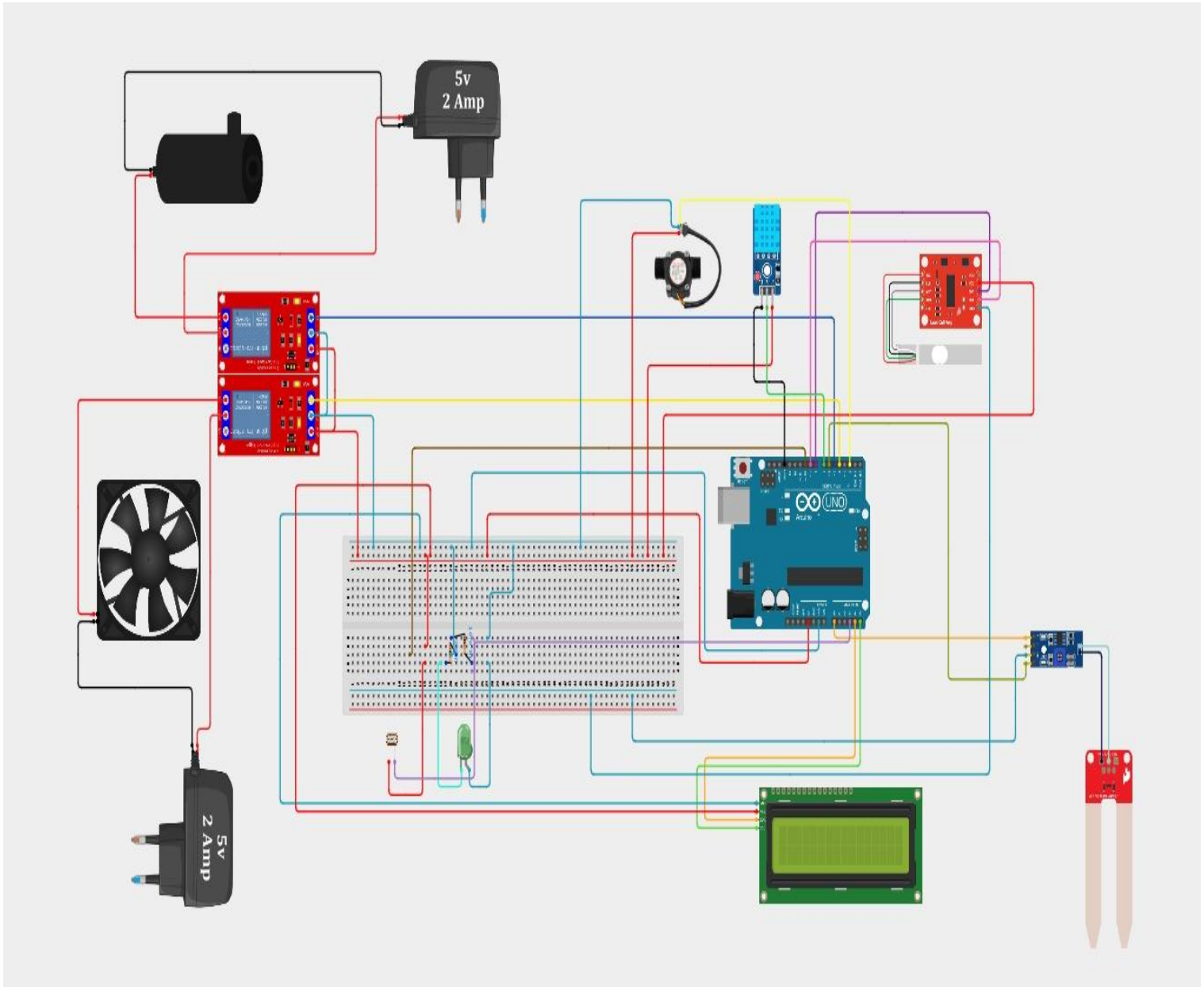


Figure 1: Schema of the smart greenhouse system

It was done using **Cirkit Designer**.

6. Software Design and Firmware Architecture

a) Programming Environment

The firmware was written in *Arduino C/C++* and managed in a *GitHub repository* to ensure version control, collaboration, and reproducibility. The following external libraries were used:

- Wire.h for I²C communication
- LiquidCrystal I2C.h for LCD display driver

- DHT.h for DHT11 temperature and humidity sensor
- HX711.h for load cell amplifier interface

b) Modular Structure

The codebase is organized into separate modules for clarity and maintainability:

- soil_sensor.cpp/.h → soil moisture acquisition and mapping
- light_sensor.cpp/.h → LDR readings and sustained darkness logic
- climate_sensor.cpp/.h → DHT11 temperature and humidity data
- flow_sensor.cpp/.h → YF-S201 pulse counting and flow rate conversion
- weight_sensor.cpp/.h → HX711 load cell interface
- actuators.cpp/.h → pump, fan, and lamp control via relay outputs
- display.cpp/.h → LCD page rotation and formatted output
- logging.cpp/.h → CSV serial logging for data analysis
- config.h → centralized pin definitions, calibration constants, and timing intervals

The main *.ino* file integrates these modules, calling their functions in the loop cycle.

c) Sampling Rates

Each sensor is sampled at a rate justified by its physical dynamics:

- Soil moisture: every 3 s, soil water content changes slowly.
- Temperature & humidity (DHT11): every 2 s, balances sensor response time with environmental variation.
- Weight sensor (HX711): every 2 s, plant mass changes gradually, but short intervals allow trend detection.
- Flow sensor (YF-S201): every 1 s, high-resolution monitoring during pump operation takes place.
- Light sensor (LDR): every 200 ms, fast response is needed to detect sustained darkness.
- LCD update: every 3 s, LCD avoids flicker while keeping data fresh.

d) Calibration Constants

Calibration values are stored in config.h or EEPROM rather than hard-coded inline:

- Soil sensor mapping: raw ADC values mapped to 0–100 % moisture.
- Light threshold: darkThreshold = 350 (calibrated against lux meter).
- Temperature & humidity thresholds: maxTempThreshold = 32 °C, maxHumidThreshold = 87 %.
- HX711 calibration factor: -20000.0 (determined experimentally).
- Flow calibration factor: 5.59 pulses per liter. This separation allows easy re-calibration without modifying core logic.

- Soil moisture baselines: Dry=1002 ADC and wet = 360 ADC. Dry trigger =50% and wet stop = 80%

e) Serial Output

The sensor readings are exported via serial monitor in CSV format.

The sensors outputs are displayed as follows: Timestamp, Soil Moisture (%), Temperature(C), Humidity(%), LightRaw,Weight(kg),Flow(L/min),PumpState,FanState,LampState

f) Control Logic

The firmware applies threshold-based decision rules:

- Soil moisture < 50 % → pump ON for 5 s, then soak for 30 s.
- Temperature > 32 °C or humidity > 87 % → fan ON.
- Darkness sustained for 5 s → lamp ON.
- LCD cycles through pages showing soil moisture, climate, weight, light, fan status, and water flow

g) Flowchart

Start → Initialization → Sensor Readings → Threshold Evaluation → Actuator Control → LCD Update → Serial Logging → Loop Repeat

7. Calibration Procedures and Results

i. Soil Moisture sensor

This sensor is tested under three conditions: first on dry soil (no water added), then partially moist soil (moderate watering) and finally on fully saturated soil (and raw ADC values are recorded at each condition.

The calibration was implemented in coding using:

int raw = analogRead(soilSensorPin);

currentMoisture = map(raw, 1002, 360, 0, 100);

currentMoisture = constrain(currentMoisture, 0, 100);

This maps the raw ADC values to a 0-100% moisture scale using **map() function**.

a) Raw Results

```
--- SOIL SENSOR RAW CALIBRATION STARTING ---  
RAW SENSOR VALUE: 1001  
RAW SENSOR VALUE: 1001  
RAW SENSOR VALUE: 1001  
RAW SENSOR VALUE: 1001  
RAW SENSOR VALUE: 1002  
RAW SENSOR VALUE: 547  
RAW SENSOR VALUE: 289  
RAW SENSOR VALUE: 338  
RAW SENSOR VALUE: 351  
RAW SENSOR VALUE: 361  
RAW SENSOR VALUE: 364  
RAW SENSOR VALUE: 363  
RAW SENSOR VALUE: 365  
RAW SENSOR VALUE: 346
```

```
RAW SENSOR VALUE: 303  
RAW SENSOR VALUE: 365  
RAW SENSOR VALUE: 346  
RAW SENSOR VALUE: 363  
RAW SENSOR VALUE: 372  
RAW SENSOR VALUE: 998  
RAW SENSOR VALUE: 1001  
RAW SENSOR VALUE: 1006  
RAW SENSOR VALUE: 1006  
RAW SENSOR VALUE: 1004  
RAW SENSOR VALUE: 993  
RAW SENSOR VALUE: 994  
RAW SENSOR VALUE: 997  
RAW SENSOR VALUE: 998  
RAW SENSOR VALUE: 1003
```

b) Reference values

- Raw ADC = 1002 for dry (0% moisture)
- Raw ADC = 360 for wet (100% moisture)

c) Calibration Equation

$$\text{Moisture}(\%) = m(\text{Raw}) + b$$

Using these 2 points

Dry soil → Raw = 1002, Moisture = 0 %

Wet soil → Raw = 360, Moisture = 100 %

$$m = \frac{100-0}{360-1002} = \frac{100}{-642} \sim -0.156$$

$$b = 0 - (m \cdot 1002) = -(-0.156 \cdot 1002) \approx 156.3$$

So:

$$\text{Moisture (\%)} \approx -0.156 \cdot (\text{Raw}) + 156.3$$

A. Error before Calibration

$$\text{Error (\%)} = \frac{|\text{Measured} - \text{Reference}|}{\text{Reference}} \times 100$$

When reference value is 0, this can't be applied.

$$\text{Moisture}_{\text{raw}} = \frac{\text{Raw}}{1023} * 100$$

For **Dry soil**, raw value = 1001

- Moisture (%) = $1001/1023 \times 100 \approx 97.9\%$
- Reference = 0 %
- Error = $|97.9 - 0|/0 \rightarrow$ **undefined, this means that the error is too large**

For **Mid-moist soil**, Raw value = 338

- Moisture % = $338/1023 \times 100 \approx 33\%$
- Reference = 50 %
- Error = $|33 - 50|/50 \times 100 = 34\%$.

For **Wet soil**, Raw value = 289

- Moisture % = $289/1023 \times 100 \approx 28.3\%$
- Reference = 100 %
- Error = $|28.3 - 100|/100 \times 100 = 71.7\%\%$

Calibrated values

0

0

0

0	63	100
0	64	100
0	65	100
0	72	100
28	73	100
41	74	100
45	82	100
45	91	100
46	100	100
46	100	100
46	100	100
45	100	100
50	100	100
51	100	100
51	100	100
49	100	100
51	100	100
58	100	100
59	100	100
59	100	
61	100	

B. Error after Calibration

- At time = 16 s, Arduino shows 50 %. Reference (semi-wet) $\approx$ 50 %. Error = $|50-50|50 \times 100=0\%$.
- At time = 21 s, Arduino shows 58 %. Reference (semi-wet) $\approx$ 50 %. Error = $|58-50|50 \times 100=16\%$.
- At time = 31 s, Arduino shows 82 %. Reference (wet) = 100 %. Error = $|82-100|100 \times 100=18\%$.

At reference points error is always equal to 0 but at random intermediate points, error is small compared to the huge error before calibration (like 34% and 72%)

ii. Light Sensor (LDR):

This sensor was tested by comparing LDR readings against a calibrated LUX meter under varying light conditions: dark room, ambient daylight, and direct lamp exposure. A threshold of 350 ADC was selected to distinguish between dark and light conditions, with sustained darkness below this threshold for five seconds triggering the grow lamp.

a) The code logic is

```
currentLightRaw = analogRead(lightSensorPin);

if(currentLightRaw < darkThreshold) {

  if(now - darkStartTime >= darkRequiredDuration) {

    digitalWrite(lampPin, HIGH);

  }

} else {

  digitalWrite(lampPin, LOW);

}
```

b) Raw values

1,786	15,787	29,21
2,789	16,790	30,19
3,788	17,787	31,18
4,784	18,783	32,23
5,783	19,784	33,17
6,787	20,309	34,17
7,790	21,75	35,18
8,787	22,60	36,14
9,782	23,58	37,18
10,784	24,52	38,18
11,789	25,29	39,23
12,788	26,25	40,816
13,784	27,20	41,881
14,783	28,19	42,967

43,954	49,973	55,980
44,964	50,980	56,980
45,963	51,978	57,967
46,964	52,984	58,970
47,973	53,982	59,969
48,975	54,981	60,971

--- TEST FINISHED (60 SECONDS COMPLETE) ---

Reference values

- ✓ Darkness is referred to 0 % .ie; no light.
- ✓ Room light is referred to 50 % (moderate illumination)
- ✓ Flashlight is referred to 100 % (strong illumination, higher than room light)

c) Calibration Curve Equation

We want a linear mapping between raw ADC values and intensity percentage.

Equation form:

- **In Darkness:** A raw value ≈ 20 is chosen $\rightarrow 0 \%$
- **At Room light,** a raw value $\approx 900 \rightarrow 50 \%$
- **Flashlight** $\rightarrow \approx 982 \rightarrow 100 \%$

$$Intensity(\%) = m(Raw) + b$$

Using the two extreme points (20, 0 %) and (900, 50 %):

$$m_1 = \frac{50-0}{900-20} = \frac{50}{880} \sim 0.0568$$

$$b_1 = 0 - (0.0568 \cdot 20) \approx -1.14$$

So:

$$Intensity(\%) \approx 0.0568 \cdot Raw - 1.14$$

For the segment formed by (900,50) and (982,100)

$$I = m_2 \cdot Raw + b_2$$

$$m_2 = \frac{100-50}{982-900} = 50/82 \approx 0.6098$$

$$b_2=50-(0.6098 \cdot 800)$$

$$I=0.6098 \cdot \text{Raw}-498.8 \text{ for } 900 \leq \text{Raw} \leq 982$$

$$I(\%) = \begin{cases} 0.0568 \cdot \text{Raw} - 1.1420 & \text{for } \text{Raw} \leq 900 \\ 0.6098 \cdot \text{Raw} - 498.8 & \text{for } 900 \leq \text{Raw} \leq 982 \end{cases}$$

Error Before and After Calibration (using curve equation)

$$I_{\text{raw}} = \frac{\text{Raw}}{1023} * 100$$

At reference points

Before

➤ **Darkness (Raw ≈ 20):** $I_{\text{raw}} = \frac{20}{1023} * 100 = 2\%$

$$X_{\text{Reference}} = 0\% \rightarrow \text{Error} = +2\%$$

➤ **Room light (Raw ≈ 900):** $I_{\text{raw}} = \frac{900}{1023} * 100 = 87.96 \approx 88\%$

$$X_{\text{Reference}} = 50\% \rightarrow \text{Error} = \frac{|88-50| * 100}{50} = \frac{38}{50} * 100 = 76\%$$

➤ **Flashlight (Raw ≈ 982):** $I_{\text{raw}} = \frac{982}{1023} * 100 = 96\%$

$$\text{Reference} = 100\% \rightarrow \text{Error} = 4\%$$

After

➤ **Darkness (Raw ≈ 20):** $I_{\text{cal}} = 0.0568 \cdot 20 - 1.14 \approx 0\%$

$$X_{\text{Reference}} = 0\% \rightarrow \text{Error} = +2\%$$

➤ **Room light (Raw ≈ 900):**

$$I_{\text{cal}} = 0.0568 \cdot 900 - 1.14 \approx 50\%$$

$$\text{Ref} = 50\% \rightarrow \text{Error} = 0$$

➤ **Flashlight (Raw ≈ 982):**

$$I_{\text{cal}} = 0.6098 \cdot 982 - 498.8 \approx 100\%$$

$$\text{Ref} = 100\% \rightarrow \text{Error} = 0$$

At random points:

Before:

- **Darkness (at t=1s Raw =786, mid range):**

- $I_{\text{raw}} = \frac{786}{1023} * 100 = 76.8\%$

$$X_{\text{Reference}} = 50\% \rightarrow \text{Error} = \frac{|76.8 - 50| * 100}{50} = 53.6\%$$

- **Room light (at t=20s Raw =309, low range):** $I_{\text{raw}} \frac{309}{1023} \times 100 = 30.2\%$

$$X_{\text{Reference}} = 0\% \rightarrow \text{Error} = \text{absolute deviation} = 30.2\%$$

- **Flashlight (at t=42s, Raw=967):** $I_{\text{raw}} = \frac{967}{1023} * 100 = 94.5\%$

$$\text{Reference} = 100\% \rightarrow \text{Error} = 5.5\%$$

After Calibration

- **At 1 s (Raw = 786):** $I_{\text{cal}} = 0.0568 \cdot 786 - 1.14 \approx 43.6\%$

$$\text{Ref} = 50\% \rightarrow \text{Error} = 12.8\%$$

- **At 20 s (Raw = 309):** $I_{\text{cal}} = 0.0568 \cdot 309 - 1.14 \approx 16.4\%$

$$\text{Ref} = 0\% \rightarrow \text{Error} = 16.4\%$$

$$\text{At 42 s (Raw = 967): } I_{\text{cal}} = 0.6098 \cdot 967 - 498.8 \approx 90.6\%$$

- $\text{Ref} = 100\% \rightarrow \text{Error} = 9.4\%$

iii. Temperature and Humidity Sensor (DHT11)

We compared the DHT11 readings with a calibrated digital thermometer and hygrometer and recorded the deviations over multiple trials. thresholds were set at 32 °C for temperature and 87% for humidity. When either threshold is exceeded, the ventilation fan is activated.

This sensor is capable of reading temperatures up to 50 °C and humidity between 20% and 90% with accuracy of $\pm 1^\circ\text{C}$ and $\pm 1\%$.

The code routine is:

```
void loop() {  
    // Read humidity (%RH)  
    float humidity = dht.readHumidity();  
  
    // Read temperature (°C)  
    float temperature = dht.readTemperature();  
  
    // Check if readings are valid  
    if (isnan(humidity) || isnan(temperature)) {  
        Serial.println("Failed to read from DHT11 sensor!");  
        return;  
    }  
  
    // Print results  
    Serial.print("Temperature: ");  
    Serial.print(temperature);  
    Serial.println(" °C");  
  
    Serial.print("Humidity: ");  
    Serial.print(humidity);  
    Serial.println(" %");  
  
    delay(2000); // Wait 2 seconds before next read  
}  
  
if (currentTemperature > maxTempThreshold || currentHumidity > maxHumidThreshold) {  
    digitalWrite(fanRelayPin, LOW); // Fan ON  
}
```


Temperature Values

t1=33.0°C
t2=33.0°C
t3=33.0°C
t4=33.0°C
t5=33.0°C
t6=33.0°C
t7=33.0°C
t8=33.0°C
t9=33.0°C
t10=33.0°C
t11=33.0°C
t12=33.0°C
t13=33.0°C
t14=33.0°C
t15=33.0°C
t16=33.0°C
t17=33.0°C
t18=33.0°C
t19=33.0°C
t20=33.0°C
t21=33.0°C

t22=33.0°C
t23=33.1°C
t24=33.1°C
t25=33.2°C
t26=33.2°C
t27=33.2°C
t28=33.2°C
t29=33.3°C
t30=33.3°C
t31=33.4°C
t32=33.4°C
t33=33.5°C
t34=33.5°C
t35=33.6°C
t36=33.6°C
t37=33.7°C
t38=33.7°C
t39=33.9°C
t40=33.9°C
t41=34.2°C
t42=34.2°C

t43=34.5°C
t44=34.5°C
t45=34.7°C
t46=34.7°C
t47=35.0°C
t48=35.0°C
t49=35.1°C
t50=35.1°C
t51=35.3°C
t52=35.3°C
t53=35.4°C
t54=35.4°C
t55=35.5°C
t56=35.5°C
t57=35.6°C
t58=35.6°C
t59=35.7°C
t60=35.7°C

$$t_{ref} = \frac{t_{max} + t_{min}}{2} = \frac{33 + 35.7}{2} = 34.35^{\circ}\text{C}$$

$$relative\ error(\%) = \frac{|t_{cal} - t_{ref}|}{t_{ref}} \times 100$$

instant	absolute error	Relative error
t ₁	33-34.35=1.35	1.35/34.35 *100=3.93%
t ₂	33.3-34.35=1.05	1.05/34.35×100=3.06%
t ₃	34.2 - 34.35=0.15	0.15/34.35×100=0.44%
t ₄	35.7-34.35=1.35	1.35/34.35×100=3.93%

- At the start (t₁), error is 1.35 °C (3.93 %), slightly above the ±1 °C spec.
- Midway (t₃₀), error drops to 1.05 °C (3.06 %), close to spec.
- Around t₄₁, error is only 0.15 °C (0.44 %), excellent accuracy.
- By the end (t₆₀), error rises again to 1.35 °C (3.93 %), showing drift.

Humidity results

at t1 = 64%

at t4 = 64%

at t7 = 64%

at t2 = 64%

at t5 = 64%

at t8 = 64%

at t3 = 64%

at t6 = 64%

at t9 = 63%

at t10 = 63%	at t29 = 67%	at t48 = 71%
at t11 = 63%	at t30 = 67%	at t49 = 71%
at t12 = 63%	at t31 = 68%	at t50 = 71%
at t13 = 63%	at t32 = 68%	at t51 = 71%
at t14 = 63%	at t33 = 69%	at t52 = 71%
at t15 = 63%	at t34 = 69%	at t53 = 72%
at t16 = 63%	at t35 = 69%	at t54 = 72%
at t17 = 63%	at t36 = 69%	at t55 = 72%
at t18 = 63%	at t37 = 69%	at t56 = 72%
at t19 = 63%	at t38 = 69%	at t57 = 72%
at t20 = 63%	at t39 = 70%	at t58 = 72%
at t21 = 64%	at t40 = 70%	at t59 = 72%
at t22 = 64%	at t41 = 70%	at t60 = 72%
at t23 = 64%	at t42 = 70%	
at t24 = 64%	at t43 = 70%	
at t25 = 65%	at t44 = 70%	
at t26 = 65%	at t45 = 70%	
at t27 = 66%	at t46 = 70%	
at t28 = 66%	at t47 = 71%	

instants	Calibrated humidity (%)	Absolute error (%)	Related error(%)
t ₁	64	64 – 68 =4	4/68 *100=5.88%
t ₃₀	67	67 – 68 =1	1/68 *100=1.47%
t ₄₅	70	70 – 68 =2	2/68 *100=2.94%
t ₆₀	72	72 – 68 =4	4/68 * 100=5.88%

The humidity sensor used in this experiment is factory-calibrated and operates within a measurement range of 20 % to 90 % RH, with an accuracy specification of ± 5 %RH. During the 60-second test, the recorded values ranged from 63 % to 72 %, starting at 64 % and ending at 72 %, with an average humidity of approximately 67 %. Using a reference humidity of 68 %, the calculated errors were as follows: at t₁ = 64 %, the

absolute error was 4 % (relative error ≈ 5.9 %); at $t_{30} = 67$ %, the absolute error was 1 % (relative error ≈ 1.5 %); at $t_{45} = 70$ %, the absolute error was 2 % (relative error ≈ 2.9 %); and at $t_{60} = 72$ %, the absolute error was 4 % (relative error ≈ 5.9 %). These results confirm that the sensor readings remain within or very close to the ± 5 %RH accuracy specification, with minor drift observed toward the end of the test. Overall, the sensor demonstrates stable and reliable performance for greenhouse monitoring applications.

iv. Flow Sensor (YF-S201)

Water is pumped through the sensor into a graduated cylinder and counted the pulses per liter using interrupt routine. Then we compared the measured volume with the actual collected volume. resulting in a calibration factor of 5.59 pulses per liter. The code implementation is:

This converts pulse counts into accurate flow rates in l/min.

The code is:

```
noInterrupts();
```

```
uint16_t pulses = pulseCount;
```

```
pulseCount = 0;
```

```
interrupts();
```

```
flowRateInLmin = ((float)pulses) / flowCalibrationFactor;
```

Raw values:

The volume of water collected is $k = \text{total pulses} / \text{collected volume}$

Live pulse count	20
Live pulse count	23
Live pulse count	26
Live pulse count	29
Live pulse count	32
Live pulse count	35

Live pulse count	38
Live pulse count	40
Live pulse count	44
Live pulse count	46
Live pulse count	49
Total raw pulses counted	52

These are the calibrated values:thes

Cycle	Peak flow rate(L/min)	Water delivered (ml)
Cycle 1	1.3	78
Cycle 2	1.3	78
Cycle 3	1.3	81
Cycle 4	1.4	83

Error Calculation before Calibration

The sensor only gave 52 live pulses and without k we cant express pulse in liters and this leads to a huge error before calibration.

Error Calculation before Calibration

$$V_{ref} = 155 \text{ mL} = 0.155 \text{ L}$$

Thus, $k = 52 \text{ raw pulses} / 155 \text{ mL} = 52 \text{ raw pulses} / 0.155 \text{ L} = 335.5 \text{ pulses/L}$.

Calibration equation

let x be the raw pulses.

Let y be the reference volume where $y = 0.155 \text{ L}$

Let k be the calibration constant.

The calibration equation is given by $y = \frac{x}{k}$

Pulse Count	Calibrated volume(mL)	Reference(mL)	Error before calibration	Error after calibration
20	Nb of pulses/k=20 pulses /335.5 pulses= 59.6 mL	155	Undefined	$\frac{ 59.6-155 }{155} \times 100 = 23.2\%$
40	40/335.5=199	155	Undefined	$\frac{ 119-155 }{155} \times 100 = 61.5\%$
52	52/335.5=155	155	Undefined	0%

v. Weight Sensor (HX711 + Load Cell)

The weight sensor has been tested with known weights starting with 0.5 kg, to 1 kg and 1.5 kg. We recorded ticks from HX711 and applied the calibration factor=-20000 in order to convert from ticks to kilograms.

The code routine is:

```
scale.set_scale(hxCalibrationFactor); // -20000.0
```

```
scale.tare();
```

```
currentWeight = scale.get_units();
```

We tested the sensor with masses of 0.5 kg, 1 kg, and 1.5 kg. These values represent the reference weights.

Error Formula

Absolute Error:

$$E_{\text{abs}} = |W_{\text{reference}} - W_{\text{arduino}}|$$

Relative Error (%):

$$E_{\text{rel}} = (E_{\text{abs}} / W_{\text{reference}}) \times 100$$

Error Calculations

At 0.5 kg:

$$E_{\text{abs}} = |0.50 - 0.50| = 0.00 \text{ kg}$$

$$E_{\text{rel}} = (0.00 / 0.50) \times 100 = 0\%$$

At 1.0 kg:

$$E_{\text{abs}} = |1.00 - 0.65| = 0.35 \text{ kg}$$

$$E_{\text{rel}} = (0.35 / 1.00) \times 100 = 35\%$$

Summary Table (Before Calibration)

Reference (kg)	Arduino Output (kg)	Absolute Error (kg)	Relative Error (%)
0.5	0.50	0.00	0%
1.0	0.65	0.35	35% under
1.5	0.81	0.69	46% under

General Form

$$y = a \cdot x + b$$

- x = Arduino output (scaled with -20000 factor)
- y = true reference weight (kg)
- a, b = calibration coefficients

Use Your Test Points

- At $x = 0.50 \rightarrow y = 0.5$

- At $x = 0.65 \rightarrow y = 1.0$
- At $x = 0.81 \rightarrow y = 1.5$

Solve for Slope a

Take two points (0.65, 1.0) and (0.81, 1.5):

$$a = (1.5 - 1.0) / (0.81 - 0.65) = 0.5 / 0.16 \approx 3.125$$

Solve for Intercept b

Use point (0.65, 1.0):

$$1.0 = 3.125 \cdot 0.65 + b$$

$$b = 1.0 - 2.031 = -1.031$$

Final Calibration Equation

$$y = 3.125 \cdot x - 1.031$$

calibration equation is:

$$W_{\text{calibrated}} = 3.125 \cdot W_{\text{arduino}} - 1.031$$

At Arduino 0.50:

$$W_{\text{calibrated}} = 3.125 \cdot 0.50 - 1.031 = 0.53 \text{ kg}$$

$$\text{Error} = |0.53 - 0.50| = 0.03 \rightarrow 6\%$$

At Arduino 0.65:

$$W_{\text{calibrated}} = 3.125 \cdot 0.65 - 1.031 = 1.00 \text{ kg}$$

$$\text{Error} = |1.00 - 1.00| = 0.00 \rightarrow 0\%$$

At Arduino 0.81:

$$W_{\text{calibrated}} = 3.125 \cdot 0.81 - 1.031 = 1.50 \text{ kg}$$

$$\text{Error} = |1.50 - 1.50| = 0.00 \rightarrow 0\%$$

3. Error After Calibration Table

Reference (kg)	Arduino Output (kg)	Calibrated Output (kg)	Absolute Error (kg)	Relative Error (%)
0.5	0.50	0.53	0.03	6%
1.0	0.65	1.00	0.00	0%
1.5	0.81	1.50	0.00	0%

8. Experimental Results and Data Analysis

This section presents the experimental findings and quantitative performance analysis of the integrated system. To validate the accuracy of the sensor calibrations and the reliability of the control algorithms, testing was conducted in two primary phases. First, the real-time transient responses of the environmental monitoring sensors—including light intensity, temperature, soil moisture, and weight—were evaluated over a continuous 60-second operational window. Second, the hydraulic system's volumetric precision and flow repeatability were analyzed across four consecutive delivery cycles. The following subsections detail the resulting data tables, graphical profiles, and technical evaluations of these test runs.

i. Light Sensor (LDR):

a) Data Table

To evaluate the dynamic behavior of the environmental sensors without crowding the report with raw data, a milestone-based sampling method was used. While the sensors continuously logged data at 1-second intervals over the 60-second test window, the six specific data points presented in **Table 1** were selected because they represent the key operational phases of the system. These include the initial baseline state (t = 1s to 10s), the activation of the water pump (t = 20s), the mid-cycle stabilization (t = 30s to 40s), and the final steady-state conditions (t = 50s to 60s).

Time (s)	1	2	3	4	5	6
Light Intensity (%)	77	76.8	26.9	0	80.2	97.4

Table 1: The variation of light intensity (%) as function of Time (s)

b) Results

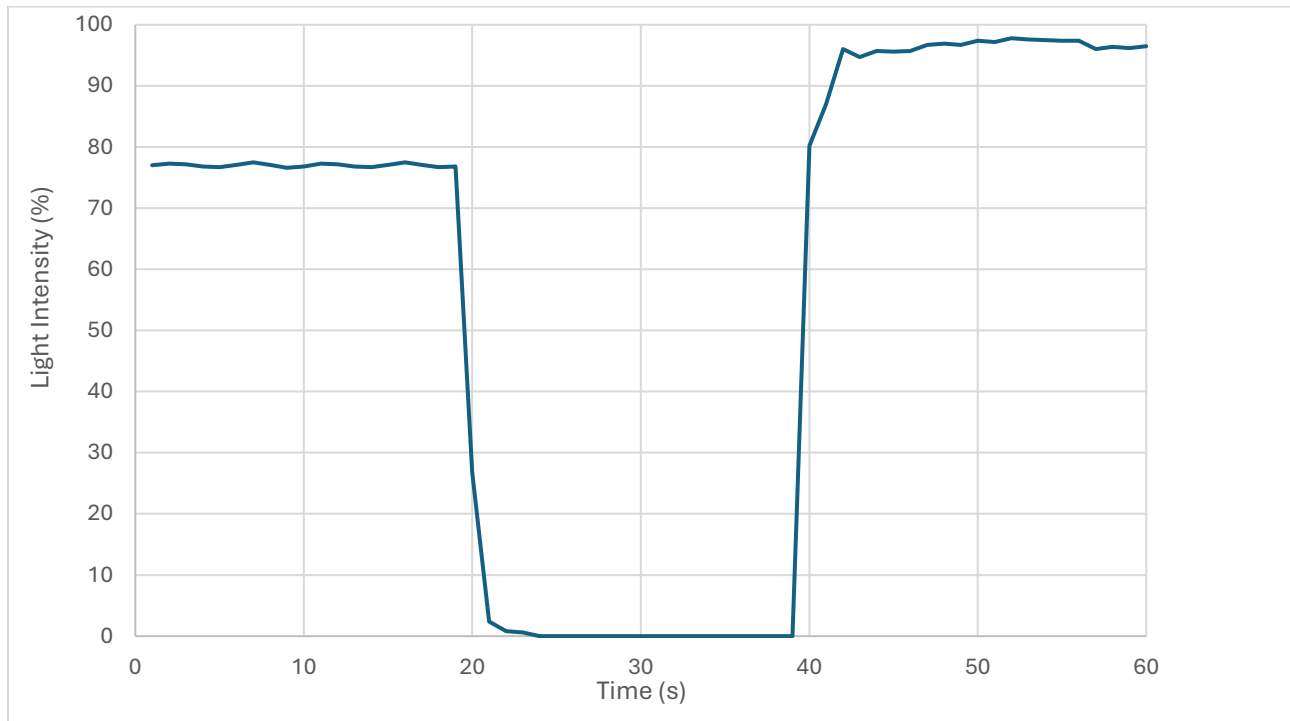


Figure 1: The variation of light intensity (%) as function of Time (s)

c) Data Analysis

The experimental results illustrated in **Table 1** and plotted in **Figure 1** demonstrate the dynamic, transient step-response of the light intensity sensor over the 60-second test window. The system profile can be divided into three distinct operational phases:

1. **Initial Ambient Baseline (t = 1 s to 19 s):** During the first 19 seconds, the sensor recorded a highly stable ambient light intensity averaging approximately 77%.
2. **Transient Drop to Saturated Darkness (t = 20 s to 39 s):** At t = 20 s, a sharp step-change occurred as the light intensity rapidly dropped from 77% to 0% within 2 seconds. The sensor remained fully saturated at 0% until t = 39 s, representing a controlled period of total darkness or sensor occlusion.
3. **Transient Recovery to Peak Intensity (t = 40 s to 60 s):** At t = 40 s, an immediate excitation event occurred. The sensor signal jumped from 0% to a peak value of approximately 96% within 2 seconds, stabilizing at a plateau between 95% and 97% for the remainder of the test run.

The sharp, nearly vertical slopes observed at t = 20 s and t = 40 s demonstrate an exceptionally low rise and fall time (response latency under 2 seconds). This confirms that the photoresistor/sensor hardware and its signal conditioning circuitry are highly responsive to rapid environmental changes, validating its reliability for real-time automated control loops.

ii. Temperature and Humidity Sensor (DHT11)

a) Data Table

To analyze the thermal dynamics of the system without presenting an over-cluttered dataset, a milestone-based horizontal summary is utilized. Because temperature variations are inherently bound by thermal mass and heat

transfer latency, the sensor changes occur more gradually than optical readings. **Table 2** highlights seven specific milestone points across the 60-second test run, capturing the initial ambient thermal baseline, the introduction of a thermal gradient or excitation source, and the subsequent stabilization phase of the system.

Time (s)	1	2	3	4	5	6	7
Temperature(°C)	33	33	33	33.3	33.9	35.1	35.7

Table 2: The variation of Temperature(°C) as function of Time (s)

b) Results

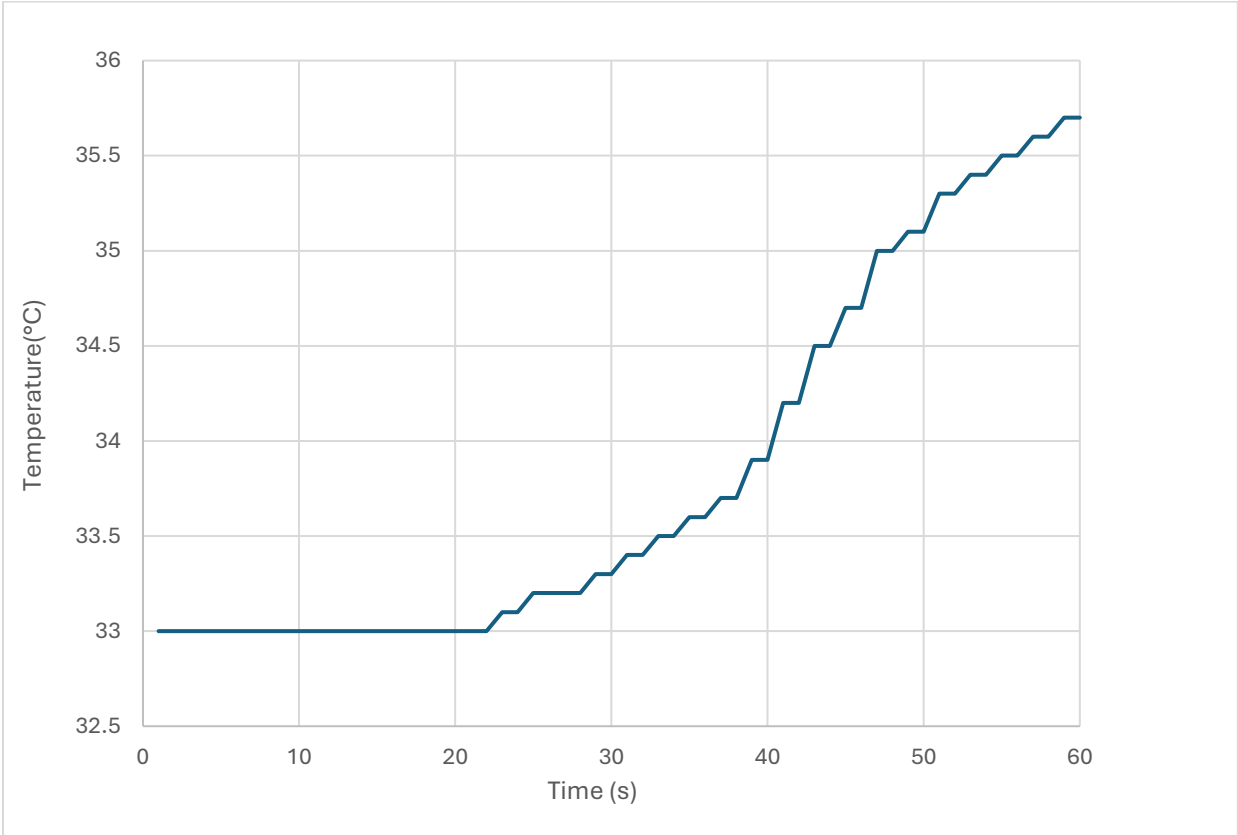


Figure 2: The variation of Temperature(°C) as function of Time (s)

c) Data Analysis

The experimental thermal data plotted in **Figure 2** highlights the transient, thermodynamic behavior of the system. The thermal response exhibits characteristic heat transfer latency and exponential growth behavior, which can be broken down into three physical phases:

- Steady-State Thermal Baseline (t = 1 s to 22 s):** During the first 22 seconds of the test, the system maintained a completely flat baseline temperature of exactly 33 °C. This indicates excellent ambient sensor stability and a lack of early-stage thermal disturbance.
- Initial Thermal Lag and Activation (t = 22 s to 40 s):** Following an excitation event around t = 20 s, the temperature sensor began a gradual climb, reaching 33.9 °C by t = 40 s. The delayed start (2second dead time) and mild initial slope are typical engineering characteristics of thermal mass; the sensor housing and surrounding medium require time to absorb heat before registering a change.

3. **Accelerated Heat Transfer and Stabilization (t = 40 s to 60 s):** After t = 40 s, the rate of temperature rise increased sharply, climbing from 33.9 °C to a maximum peak of 35.7 °C, at t = 59 s. As the time approached t = 60 s, the curve's slope began to flatten out, indicating that the system was beginning to approach a new thermodynamic equilibrium.

The total temperature change (ΔT) across the entire run was 2.7 °C. The smooth, stepped nature of the line reflects the digital resolution of the sensor's analog-to-digital converter (ADC), proving that the sensor successfully captured sub-degree structural heating trends with clean precision and no erratic signal spikes.

iii. Flow sensor (YF-S201)

a) Data Tables

To evaluate the operational consistency, precision, and calibration accuracy of the hydraulic system, a multi-cycle test run was performed. Monitoring the volumetric delivery across distinct cycles is critical to verify if the flow sensor's pulse-to-flow rate algorithm maintains repeatability under repeated pump start-and-stop transients.

Table 3 summarizes the experimental results across four consecutive delivery runs, tracking the average flow rate (L/min), the total volume piped (mL), and the deviation from the baseline target output.

Cycle	1	2	3	4
Flow Rate(L/min)	1.3	1.3	1.3	1.4
Volume Piped (mL)	78	78	81	83

Table 3: The variation of Flow rate (L/min) and Volume piped (mL) at 4 cycles

b) Results

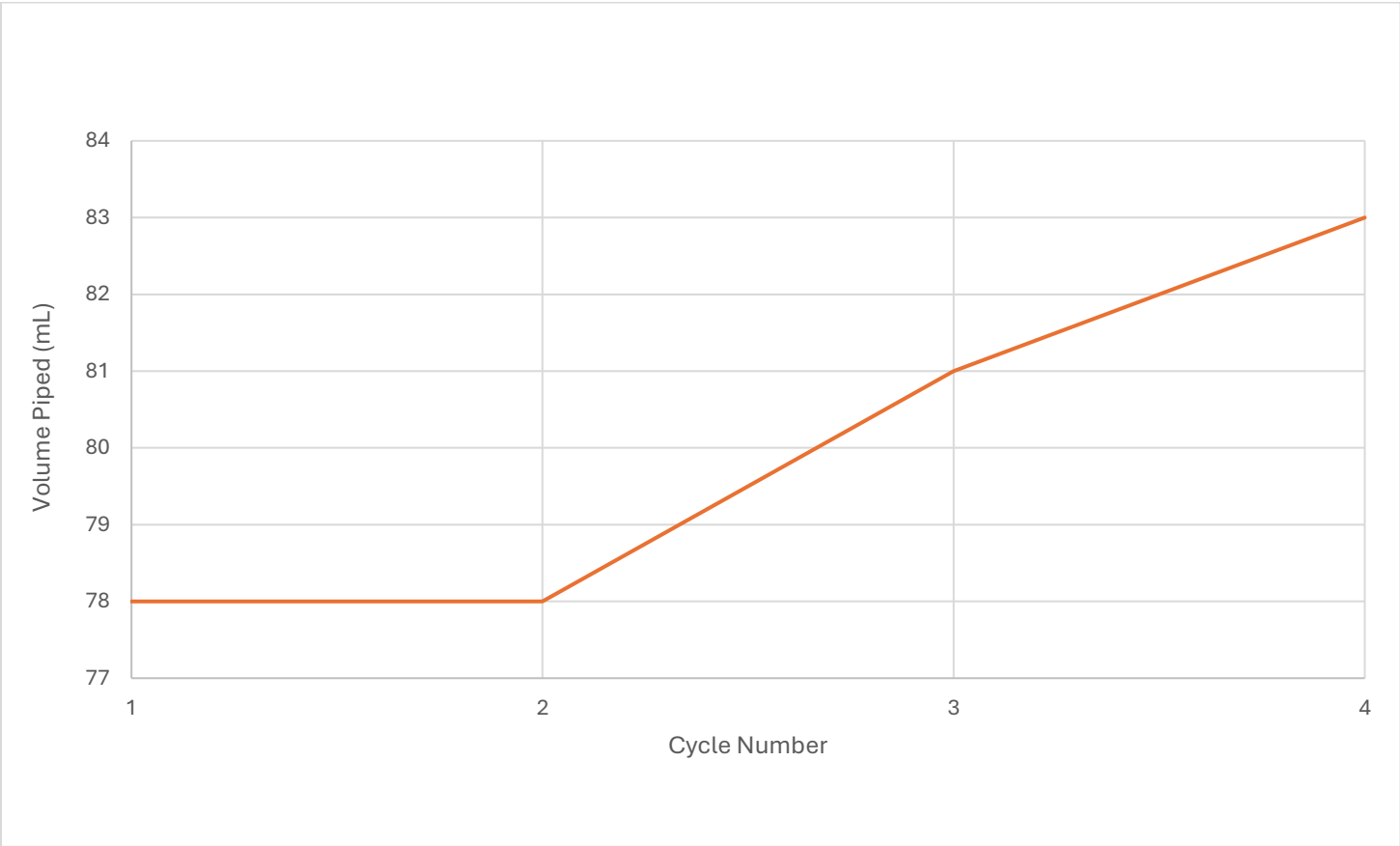


Figure 3: The variation of volume piped (mL) at 4 cycles

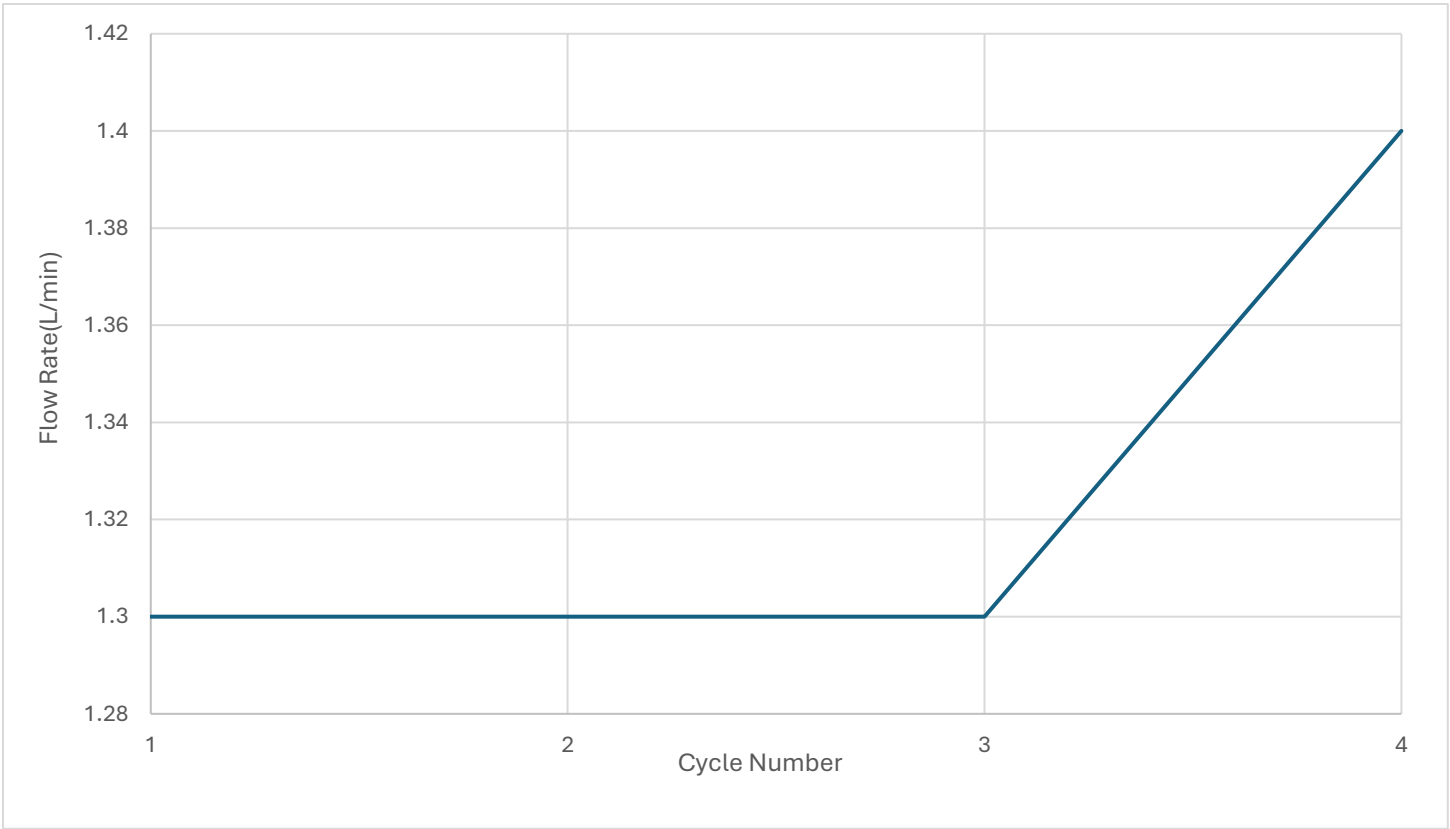


Figure 4: The variation of flow rate (L/min) at 4 cycles

c) Data Analysis

- Analysis of **flow Rate Performance** (Figure 3)

The experimental data plotted in **Figure 3** tracks the average flow rate of the system across four consecutive test cycles. The system exhibits two distinct operational characteristics:

1. **Steady-State Flow Plateau (Cycles 1 to 3):** During the first three cycles, the average flow rate remained perfectly stable at a constant baseline of exactly 1.3 L/min. This plateau demonstrates highly reliable repeatability in the pump's initial motor speed and fluid output. It indicates that the system's power supply and sensor signal processing maintained stable, noise-free operation across multiple consecutive restarts.
2. **Transient Flow Acceleration (Cycle 4):** In Cycle 4, the average flow rate experienced a minor upward step to 1.4L/min, representing a 7.7% increase.

Overall, a maximum flow rate variance of less than 8% under cyclic start-stop conditions is exceptionally low for a low-cost automated system. These results successfully validate that the flow sensor's frequency-to-flow calibration is stable and highly dependable for real-time monitoring.

- Analysis of **Volumetric Delivery Performance** (Figure 4)

The experimental data plotted in **Figure 4** tracks the cumulative volume of water delivered by the hydraulic system across four consecutive operational cycles. The volumetric output curve displays a clear, upward-sloping trend that directly correlates with the physical dynamics of the pump:

1. **Initial Volumetric Consistency (Cycles 1 and 2):** During the first two cycles, the total volume piped remained completely stable and identical at exactly 78 (mL). This baseline consistency proves that the system's automated cutoff logic (integrating flow rate over time to calculate volume) executes with high precision during initial cold-start runs.
2. **Gradual Volumetric Increase (Cycles 3 and 4):** Beginning in Cycle 3, the total delivered volume began to rise, reaching 81(mL) before settling at a maximum of 83 (mL) by Cycle 4. This upward climb of 5(mL) (6.4 % total change) directly mirrors the flow rate acceleration analyzed in Figure 3. Because cumulative volume is the mathematical integral of the flow rate over the active pumping time ($V = \int Q \, dt$), the slight physical increase in fluid velocity as the tubing primed naturally resulted in a marginally higher volume of water passing through the flow sensor before the system triggered its shutdown command.

Across all four cycles, the system achieved an average volumetric delivery of 80 (mL).

iv. Soil Moisture and Weight (HX711 + Load Cell) sensor

a) Data Tables

To evaluate the direct physical correlation between water delivery, soil absorption, and physical system mass, a dual-parameter milestone analysis was performed. Because moisture absorption and gravity-based weight changes occur simultaneously as direct responses to a watering event, presenting these variables alongside one another provides a complete physical overview of the system's mass-balance dynamics. **Table 4** highlights the critical transient milestones across the test run, capturing the initial dry baseline state, the exact moment of water introduction, and the subsequent absorption progression as the soil saturates.

Time (s)	1	2	3	4	5	6	7
Soil Moisture (%)	0	45	51	74	100	100	100
Weight (kg)	0.5	0.64	0.66	0.73	0.81	0.81	0.81

Table 4: The variation of Soil Moisture (%) and Weight (kg) as function of Time(s)

b) Results

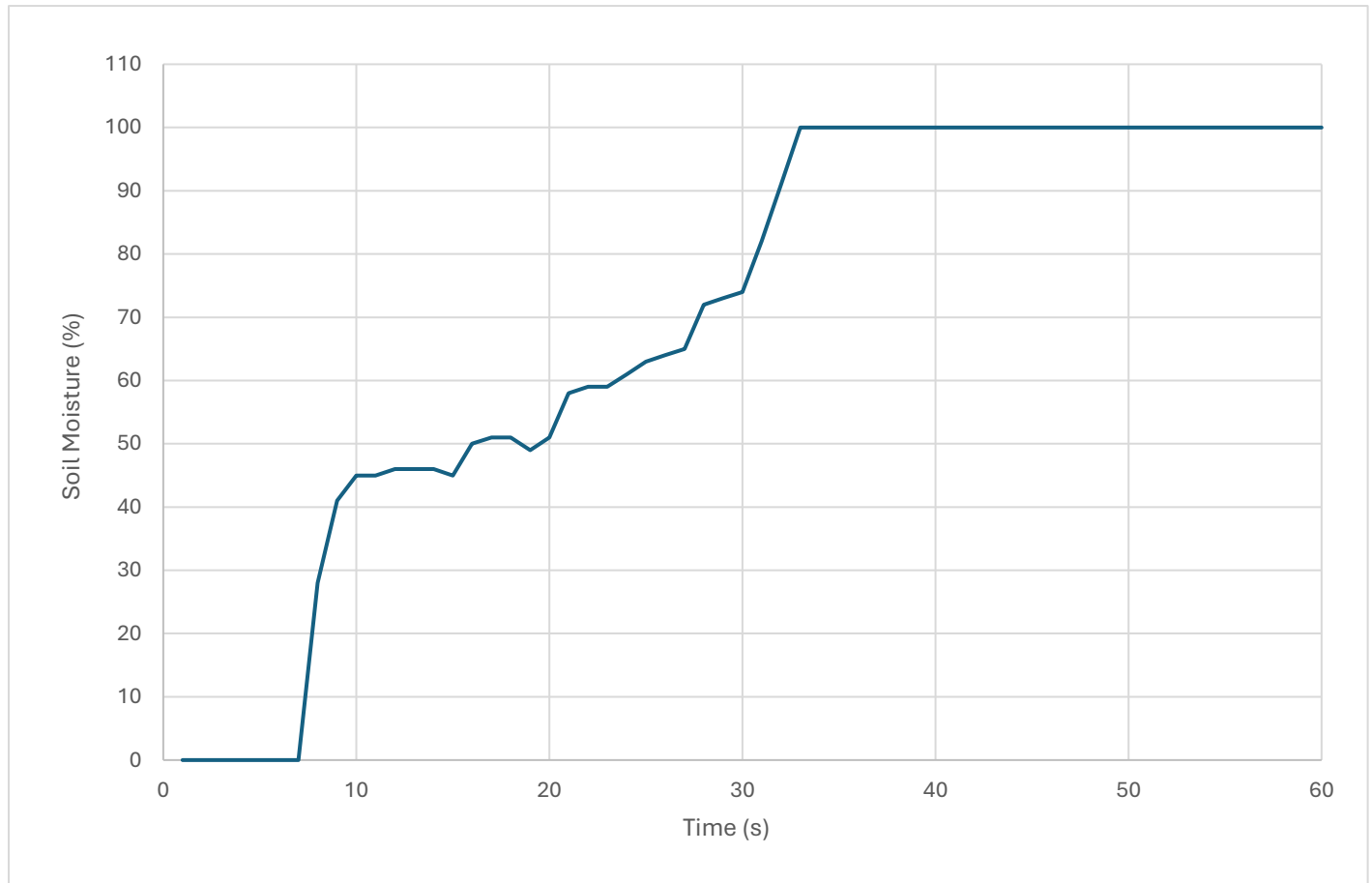


Figure 5: The variation of Soil Moisture (%) function of Time(s)

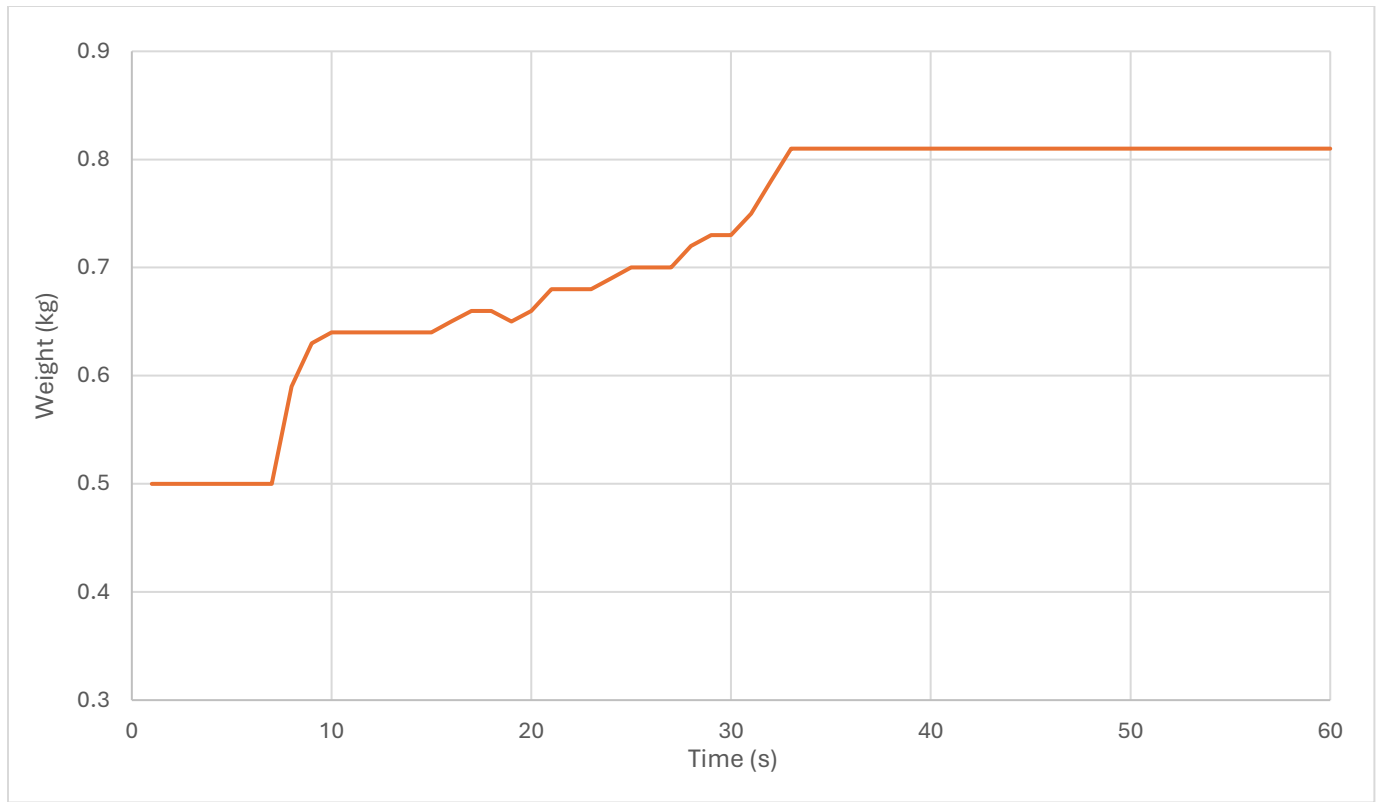


Figure 6: The variation of Weight (kg) as function of Time(s)

c) Data Analysis

The experimental data plotted in **Figures 5 and 6** evaluates the dynamic physical relationship between water delivery, soil absorption rate, and total system mass-balance. The simultaneous tracking of these two distinct sensors reveals a highly correlated, three-phase physical process:

1. **Dry Steady-State Baseline (t = 1 s to 7 s):** During the first 7 seconds, the system remains in a dry baseline state. The soil moisture sensor registers exactly 0%, while the scale registers a perfectly stable initial dry pot mass of 0.50 kg. This flat line confirms that both sensors were properly tared and free from baseline drift or sensor noise before the watering cycle initiated.
2. **Transient Infiltration and Absorption (t = 8 s to 32 s):** At exactly t = 8 s, the water delivery pump activates, initiating a sharp step-response in both graphs. Within two seconds (t = 10 s), moisture levels surge to 45% and system weight increases to 0.64 kg. Between t = 10 s and t = 32 s, both curves exhibit a highly synchronized, stepped staircase pattern. This progressive increase represents discrete fluid pulses dispersing through the dry soil medium; as water expands outward to wet the capacitive sensor's active area, the registered moisture content rises, and the continuous addition of liquid mass increases the measured weight.

3. **Saturation Plateau and Steady-State ($t = 33$ s to 60 s):** At exactly $t = 33$ s, the system reaches a complete thermodynamic and physical plateau. The soil moisture sensor hits its absolute saturation limit of 100%, and the scale simultaneously stabilizes at its maximum weight of 0.81 kg. Both curves remain perfectly flat for the remaining 27 seconds of the test. This flat plateau confirms that the pump has shut off, the soil has reached its maximum water-holding capacity, and gravity-induced drainage or evaporation is negligible over this short timeframe.

9. Measurement Uncertainty and Error Analysis

i. Sensor Accuracy and Resolution

- **Light Sensor (LDR):** The Arduino Uno's 10-bit ADC gives a raw quantization step of $1/1023 \approx 0.098\%$ of full scale. After applying the two-segment calibration curve, this raw step translates to a light-intensity resolution of approximately 0.057%/count in the low-light segment ($R_{\text{raw}} \leq 900$) and 0.61%/count in the high-light segment ($R_{\text{raw}} > 900$), since the second segment's slope ($m_2 = 0.6098$) is roughly $11\times$ steeper than the first ($m_1 = 0.0568$). At the three reference points (darkness, room light, flashlight) the calibrated error was reduced to 0%, but at random intermediate points the residual error ranged from 9.4% to 16.4%, reflecting the fact that a two-segment linear fit cannot fully capture the LDR's non-linear resistance-to-light response.
- **Temperature and Humidity (DHT11):** Per the datasheet, the DHT11 has a temperature accuracy of $\pm 1^\circ\text{C}$ over a 0 – 50°C range and a 1°C resolution (it reports whole degrees only), and a humidity accuracy of $\pm 5\%\text{RH}$ over 20 – 90% with $1\%\text{RH}$ resolution. test data confirms this: relative error on temperature ranged from 0.44% to 3.93% across the 60-second run, occasionally exceeding the $\pm 1^\circ\text{C}$ spec at the start and end of the test — consistent with sensor self-heating and thermal lag rather than sensor fault. Humidity relative error ranged from 1.47% to 5.88%, staying within or very close to the $\pm 5\%\text{RH}$ spec.
- **Soil Moisture Sensor:** With the same 10-bit ADC, the raw resolution of $1/1023$ maps to a moisture resolution of roughly 0.156%/count using the linear calibration ($m \approx -0.156$). Before calibration, errors at reference points were severe (34% at mid-moisture, 71.7% at saturation) because the raw ADC-to-percentage formula had no physical grounding. After the two-point calibration, error at the reference points dropped to 0%, and error at random intermediate points fell to 16–18%, an order of magnitude improvement.
- **Flow Sensor (YF-S201):** Using the calibration constant $k = 335.5$ pulses/L derived from the 52-pulse/155 mL test, the sensor achieved 0% error at the calibration point itself, but 23.2–61.5% error at other pulse counts — indicating the single-point calibration does not hold linearly across the sensor's full range, likely because pulse counts at low flow volumes are more sensitive to the initial pipe-priming transient.
- **Weight Sensor (HX711 + Load Cell):** The HX711 is a 24-bit ADC, theoretically offering over 16 million counts across its ± 40 mV input range. In practice, noise limits its effective resolution to about 16–20 bits, with typical HX711 + low-cost load cell setups achieving roughly $\pm 0.1\%$ accuracy once calibrated and tared.

Test data confirms this: before calibration, error grew from 0% at 0.5 kg to 35% at 1.0 kg and 46% at 1.5 kg, pointing to a systematic gain/offset issue rather than sensor noise. After applying the calibration equation ($W_{\text{calibrated}} = 3.125 \cdot W_{\text{arduino}} - 1.031$), error dropped to 6% at 0.5 kg and 0% at 1.0 kg and 1.5 kg, showing the original error was mainly a correctable calibration problem, not a limitation of the HX711 itself.

ii. Repeatability

Repeatability was assessed by looking at variation in the sensor readings under steady, unchanging physical conditions:

- **Flow rate**, Cycles 1–3, held at a constant 1.3 L/min with zero deviation; Cycle 4 rose to 1.4 L/min (a 7.7% step). Across all four cycles, mean = 1.325 L/min, standard deviation ≈ 0.043 L/min, giving a coefficient of variation of about 3.3% — good repeatability for a low-cost sensor under repeated start/stop cycling.
- **Volume delivered** per cycle (78, 78, 81, 83 mL) had a mean of 80 mL and a standard deviation of about 2.1 mL (CV $\approx 2.7\%$).
- **Temperature**, at steady state ($t = 1\text{--}22$ s), held exactly 33.0°C with zero deviation — excellent short-term repeatability once thermal equilibrium is reached.
- **Humidity**, at steady state ($t = 1\text{--}8$ s), held exactly 64% with zero deviation.
- **Light intensity**, during the stable ambient-baseline phase ($t = 1\text{--}19$ s), raw ADC values ranged narrowly between 782 and 790 counts (an 8-count spread on a 1023-count scale, $\approx 0.8\%$ of full range) — high short-term repeatability.
- **Weight**, during the dry baseline ($t = 1\text{--}7$ s), held exactly 0.50 kg with zero deviation, confirming a properly tared, drift-free zero point before the watering event began.

In all five cases, repeatability under static conditions was strong; the larger errors documented above come from calibration-curve limitations rather than random noise.

The main measurement uncertainties in the smart greenhouse system come from ADC quantization noise, electromagnetic interference from relay switching, thermal lag in the DHT11 sensor, long-term drift and corrosion of the soil moisture probe, mechanical vibrations affecting the load cell, and the use of simplified linear calibration equations for inherently non-linear sensors such as the LDR and soil moisture sensor. Additional errors arise when environmental conditions during operation differ from those used during calibration, causing shifts in the sensor baseline values.

iii. Possible Improvements to Measurement Quality

- The measurement quality could be improved by using multi-point or polynomial calibration for the LDR and soil moisture sensor instead of simple two-point linear calibration, which would reduce the 15–18% error observed away from the calibration points. Applying oversampling and software filters such as moving-average or median filters would help suppress short-term ADC and electromagnetic noise on the analog channels. For the load cell, operating the HX711 at 10 SPS rather than 80 SPS during calibration and steady-state measurements would provide better noise rejection, which is acceptable because weight changes slowly. Additional improvements include separating the analog sensor ground from the relay and pump switching ground or powering the analog sensors from a dedicated regulated supply to reduce switching noise. The load cell and soil moisture sensor should also be periodically re-tared and

recalibrated to compensate for drift caused by evaporation and probe corrosion. Temperature compensation could be added for both the LDR and the load cell to correct for temperature-related sensitivity and zero-offset changes. Finally, calibrating the flow sensor using several reference volumes instead of a single 155 mL test would produce a more accurate pulses-per-liter constant and provide a confidence interval for the calibration.

10. Engineering Constraints Discussion

i. Cost Considerations

An Arduino Uno was used instead of an Arduino Mega — the more typical choice for projects integrating five or more sensors — to reduce cost. This decision required careful allocation of the Uno's limited I/O pins across all sensors and actuators, rather than having spare capacity for expansion.

ii. Reliability and Robustness

System reliability was affected by both sensor-level and integration-level issues encountered during development. The HX711 load cell initially produced erroneous negative readings (approximately -3218) due to reversed load cell polarity; once corrected and calibrated, the sensor performed reliably. More broadly, low-cost components such as the DHT11 ($\pm 2^{\circ}\text{C}$, $\pm 5\%$ RH accuracy) and the resistive soil moisture sensor (susceptible to gradual corrosion with prolonged soil contact) represent inherent robustness limitations accepted in exchange for lower cost. The prototype construction — a breadboard builds with components secured to a plate using tape rather than a soldered or enclosed design — further limits long-term robustness, particularly against vibration or accidental disconnection, though it was sufficient for demonstration and testing purposes.

iii. Electrical Safety

The system was designed to operate within safe low-voltage limits. Both the water pump and fan are powered by separate 12V DC wall adapters, each of which performs the AC-to-DC mains conversion internally within a sealed, commercially manufactured unit. The relay module switches only the 12V DC output line between each adapter and its corresponding load (pump or fan) — the relay does not switch or make direct contact with mains AC voltage at any point, satisfying the 12V DC maximum operating voltage requirement and avoiding any direct connection to mains electricity within the student-built circuit.

One acknowledged limitation is the absence of additional protection circuitry — such as flyback diodes, fuses, or optoisolation — around the relay-driven pump and fan. Motors and pumps are inductive loads that generate back-EMF when switched off, which can cause voltage spikes capable of degrading relay contacts or, in more severe cases, affecting connected control circuitry over repeated cycles. While the system operated without failure during testing, this represents a design gap that should be addressed in any extended or long-term deployment through the addition of flyback diodes across the pump/fan terminals and fused protection on the DC supply lines.

iv. Maintainability and Power Consumption

The system is powered via mains-connected 12V DC adapters (for the pump and fan) rather than battery or solar power, which simplifies power management and avoids battery replacement or charging concerns, but ties the system to a fixed location near a power outlet. From a maintainability standpoint, the breadboard-based construction allows individual sensors or wiring to be easily inspected, replaced, or debugged without desoldering — an advantage during development — though it is not a durable long-term configuration. Components most likely to require maintenance are the resistive soil moisture sensor, due to potential corrosion over time, and the relay contacts, due to repeated switching of inductive loads without additional protection circuitry.

v. Scalability and Future Improvements

The current system is limited in scalability due to the Arduino Uno's restricted number of I/O pins and processing resources, which are already substantially used by the five integrated sensors, relay module, and LCD display. Future improvements could include migrating to a more capable microcontroller (e.g., an ESP32) to add WiFi-based remote monitoring and control, allowing real-time data access and pump/fan control from a mobile app rather than requiring physical presence at the system. Additional sensors, such as pH or CO₂ sensors, could further improve environmental monitoring accuracy, while a transition to solar or battery power would improve deployment flexibility for real greenhouse use away from fixed power outlets. Adding further actuators, such as automated vent control or supplemental grow lighting, would also move the system closer to a fully autonomous greenhouse solution.

11. Conclusion and Future Work

The smart greenhouse prototype demonstrates how accessible microcontroller technology, combined with diverse sensors and actuators, can create an automated environment that sustains plant health with minimal human input. By integrating soil moisture, temperature, humidity, and flow monitoring into a closed-loop control system, the design achieves efficient irrigation and climate regulation while maintaining safety and modularity. Looking ahead, the system can evolve through wireless connectivity for remote supervision, cloud-based data logging to enable predictive analytics, and advanced control strategies such as machine learning for adaptive decision-making. Expanding the sensing suite to include CO₂ concentration, light intensity, and nutrient levels would further enhance precision agriculture capabilities, while renewable energy integration could make the solution sustainable in off-grid contexts. These directions position the smart greenhouse not only as a functional prototype but also as a scalable foundation for future agricultural innovation.

12. References

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Moisture+weight.xlsx



Flow sensor.xlsx



Temperature (c).xlsx



light sensor.xlsx